



Assessment of cardiac output as a predictor of metabolic rate in rainbow trout

J. C. BRODEUR*, D. G. DIXON AND R. S. MCKINLEY

Department of Biology, University of Waterloo, Waterloo, Ontario N2L 3G1, Canada

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When water temperature was increased from 12 to 27°C at a rate of 2°C h⁻¹, oxygen consumption of rainbow trout *Oncorhynchus mykiss* was correlated strongly with both heart rate and blood oxygen extraction but the relationship with cardiac output was variable and weak. On the other hand, when water temperature was decreased from 21 to 12°C at a rate of 0.5°C h⁻¹, oxygen consumption was correlated with both heart rate and cardiac output but not with blood oxygen extraction. When fish were forced to swim increasingly faster, heart rate, cardiac output and blood oxygen extraction all correlated positively with oxygen consumption. For both cardiac output and heart rate, the slope of the regression line with oxygen consumption was elevated significantly more when the fish were forced to swim at increasingly higher swimming speeds than when water temperature was increased or decreased. The variation of the regression lines between cardiac output and oxygen consumption indicated that cardiac output presents few advantages over heart rate as a predictor of metabolic rate.

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Key words: cardiac output; metabolic rate; temperature; heart rate; swimming; rainbow trout.

INTRODUCTION

The increase in popularity of bioenergetic modelling to address fisheries research problems has accentuated greatly the need for information on the metabolic rate of free-living fish. Field estimates of metabolism are difficult to obtain in fish because the doubly-labelled water technique usually used in terrestrial animals is unsuitable due to the high water turnover rates of fish. Currently, the only way of estimating field metabolic rates of free-living fish is through the telemetry of physiological correlates of oxygen consumption such as ventilation rate (Rogers & Weatherley, 1983), tailbeat frequency (Young *et al.*, 1972; Stasko & Horrall, 1976), locomotor muscle electromyograms (EMGs) (Weatherley *et al.*, 1982; Hinch *et al.*, 1996; Briggs & Post, 1997) and heart rate (Priede & Young, 1977; Lucas *et al.*, 1991). However, most parameters so far measured have some limitations. For example, EMG telemetry can be used to estimate the metabolic cost of activity, but does not allow prediction of the influence of other factors such as feeding or environmental stress. Heart rate, although more likely to integrate all respiratory components, has been criticized also as an indicator of metabolism because its relationship to oxygen consumption can be described by numerous curves, depending on the physiological state of the fish and on environmental factors (Thorarensen *et al.*, 1996a).

*Author to whom correspondence should be addressed at present address: Gatty Marine Laboratory, School of Environmental and Evolutionary Biology, University of St Andrews, St Andrews, Fife, Scotland, KY16 8LB, U.K. Tel.: 01334 463 449; fax: 01334 463 443; email: jb45@st-and.ac.uk

This variation in the relationship between heart rate and oxygen consumption results from the fact that heart rate is only one of several cardiovascular variables that can be modified in response to increased tissue oxygen demand. This is shown by the Fick equation which relates oxygen consumption to the cardiovascular variables:

$$V_{O_2} = f_H \cdot V_H \cdot E_{O_2}$$

where V_{O_2} is oxygen consumption, f_H is heart rate, V_H is stroke volume, and E_{O_2} is blood oxygen extraction, i.e. the amount of O_2 that is extracted from the blood as it passes through the tissues.

Thorarensen *et al.* (1996a) suggested that more accurate estimates of metabolic rate could be obtained by monitoring cardiac output (the product of heart rate and stroke volume) rather than heart rate. Recently Webber *et al.* (1998) have supported this view by showing the presence of a strong relationship between cardiac output and oxygen consumption in the cod *Gadus morhua* L. This relationship was established by forcing the fish to swim at various speeds and taking simultaneous measurements of cardiac output and oxygen consumption over several weeks. However, if cardiac output is to be measured telemetrically and used as an indicator of metabolic rate in wild fish, it is essential that the calibration for oxygen consumption be done rapidly without holding the animal for extended periods. Furthermore, it should be established whether cardiac output is a good indicator of metabolic rate when factors other than swimming modify fish metabolism, since this would represent the main advantage of a telemetric device measuring cardiac output over the currently available EMG transmitters.

The objective of the present study was to determine whether cardiac output is a better correlate of oxygen consumption than heart rate when the metabolic rate of rainbow trout *Oncorhynchus mykiss* (Walbaum) is modified either by an acute increase or decrease in water temperature or by an increase in swimming speed. An effort was made to keep the procedures simple so that they could be used as a protocol to calibrate an eventual telemetric device for wild animals.

MATERIALS AND METHODS

EXPERIMENTAL ANIMALS

Rainbow trout were obtained from Rainbow Springs Hatchery (Thamesford, Ontario) and held at 12°C in a 2 × 2 m tank. Average body weight and total length of the fish were 396 ± 45 g and 32.6 ± 1.3 cm (mean ± S.D.). Fish were starved for at least 48 h before use to avoid the influence of digestive state on cardiac output and oxygen consumption. New animals were used for each treatment.

SURGICAL PROCEDURES

Anaesthesia was induced by immersing the fish in an aqueous solution of clove oil (120 ml l⁻¹) as described by Anderson *et al.* (1997). Fish were maintained under anaesthesia during surgery by irrigating the gills continuously with a 30 mg l⁻¹ solution of clove oil. A Doppler flow probe was installed around the ventral aorta of the fish to allow blood velocity to be measured and cardiac output, heart rate, and stroke volume to be calculated from these data. To install the Doppler flow probe, the operculum and gills were lifted gently and held in an open position with a specially designed plastic pad.

Connective tissue covering the ventral aorta was teased away carefully on a section c. 4 mm long, and a cuff-type Doppler flow probe (Iowa Doppler Products, Iowa City, IA, 20 MHz) was placed around the blood vessel. The cuffs were selected to match the diameter of the vessel. The lead wire from the probe was sutured to the skin of the fish at one location on the edge of the opercular cavity and at three locations on the body wall. The surgery lasted c. 15 min.

RESPIROMETRY

Immediately after surgery, fish were placed in a 120-l blazka-type respirometer and maintained at low swimming speed (0.5 m s^{-1}) overnight to allow recovery from surgery and acclimation to the respirometer. Water was pumped from a 400-l reservoir through the respirometer, and back into the reservoir. The reservoir was oxygenated and supplied continuously with an inflow of fresh water. Water temperature in the respirometer was 12°C .

Dissolved oxygen concentrations in the respirometer were measured in a parallel external circuit where water was pumped out of the respirometer and brought back into it after flowing over the electrode of a YSI oxygen meter (Yellow Springs, OH). Fish oxygen consumption was estimated by sealing the respirometer (by shutting valves on the inflow and outflow tubes) and measuring the decrease in dissolved oxygen concentration over either 15 (swimming experiment) or 30 (temperature experiments) min. Oxygen consumption (V_{O_2}) was adjusted to a standard body mass of 1 kg using the mass exponent of 0.8 (Saunders, 1963):

$$V_{\text{O}_2(1 \text{ kg})} = M^{-0.8} V_{\text{O}_2}$$

where M is body mass.

CARDIAC MEASUREMENTS

To determine the relationship between oxygen consumption and the cardiovascular parameters, blood velocity in the ventral aorta was sampled continuously with a Doppler flow probe over the 15 min (swimming experiment) or 30 min (temperature experiments) during which oxygen consumption was measured. Doppler signals were recorded at 20 Hz using a pulsed Doppler flowmeter (545C-4, Department of Bioengineering, University of Iowa, IA). Cardiac output, heart rate and stroke volume were calculated from the Doppler signal for each 8 s of recording, and the average value for the 15 or 30 min recording was calculated for each parameter.

At the end of the experiment, an *in situ* calibration of the Doppler flow probe was performed on each fish to obtain absolute values of blood flow. After exposing the heart, the bulbus arteriosus was catheterized so that the ventral aorta could be perfused. Then an infusion pump (model 70-2000, Harvard Apparatus, Holliston, MA) was used to perfuse calf blood through the ventral aorta at a range of flow rates while the Doppler signal was recorded. The linear relationship obtained between the flow rate in ml min^{-1} and the Doppler signal in volts was used to transform the Doppler signals to absolute blood flow.

Blood oxygen extraction (E_{O_2}) was calculated from cardiac output (C_{O}) and oxygen consumption (V_{O_2}) using a modified version of the Fick equation:

$$E_{\text{O}_2} = V_{\text{O}_2} \cdot C_{\text{O}}^{-1}$$

The contribution of cardiac output and blood oxygen extraction to the variations in oxygen consumption was calculated as the log of the factorial increase of the variable over the log of the factorial increase of oxygen consumption, e.g. for cardiac output:

$$100 \log[C_{\text{O}(2)} \cdot C_{\text{O}(1)}] \log[V_{\text{O}_2(2)} \cdot V_{\text{O}_2(1)}^{-1}]^{-1}$$

where (2) denotes the higher values and (1) denotes the lower values of C_{O} and V_{O_2} .

EXPERIMENTAL PROTOCOL

Increasing water temperature

After the fish had been allowed to recover from surgery overnight at 12°C, the water reservoir supplying the respirometer was heated at a rate of 2°C h⁻¹. Oxygen consumption and Doppler signal were measured simultaneously over a 30 min period when water temperature reached 12, 15, 18, 21, 24 and 27°C. Swimming speed was maintained at 0.5 m s⁻¹ throughout the experiment.

Decreasing water temperature

After introduction of the fish in the respirometer, the water of the reservoir was heated to 21°C overnight at a rate of 0.5°C h⁻¹. On the next morning, the water temperature was reduced at a rate of 0.5°C h⁻¹ and measurements of oxygen consumption and Doppler signal were taken simultaneously over 30 min when water temperature reached 21, 18, 15 and 12°C. Swimming speed was maintained at 0.5 m s⁻¹ during the experiment.

Increased swimming speed

On the morning following the surgery, swimming speed was increased by 0.33 m s⁻¹ every 30 min up to 1.48 m s⁻¹. Oxygen consumption and Doppler signal were measured simultaneously over the first 15 min the fish spent swimming at each speed. During the remaining 15 min, fresh oxygenated water was allowed to recirculate in the respirometer to avoid a depletion of dissolved oxygen concentrations.

STATISTICAL ANALYSIS

Relationships between the different parameters measured were estimated using linear or exponential regressions, depending on the nature of the relationship. Regression lines were compared using an analysis of covariance.

RESULTS

RESPONSE TO AN INCREASE IN WATER TEMPERATURE

An exponential increase of oxygen consumption (V_{O_2}) from 1.9 to 5.0 mg kg⁻¹ min⁻¹ was observed when water temperature (T) was elevated from 12 to 27°C at a rate of 2°C h⁻¹ [Fig. 1(a); $V_{O_2} = 0.724 \times 1.073^T$, $r^2 = 0.588$, $P < 0.001$, $n = 9$]. The increase of blood oxygen extraction from 0.07 to 0.13 mg ml⁻¹ contributed most ($62.4 \pm 15.5\%$, mean $\pm$ s.e.) to the elevation of oxygen consumption [Fig. 1(a)]. The small increase of cardiac output from 30.9 to 42.7 ml min⁻¹ kg⁻¹ accounted for the rest of the increase in oxygen consumption [Fig. 1(a)]. The increase of cardiac output was only moderate because a decrease in stroke volume was associated with the large elevation of heart rate from 52.7 to 110.3 beats min⁻¹ [Fig. 1(b)]. Cardiac output (C_O) was correlated only weakly with oxygen consumption [Fig. 2(a); $V_{O_2} = 0.052 C_O + 1.246$, $r^2 = 0.132$, $P = 0.01$, $n = 9$], while both blood oxygen extraction (E_{O_2}) and heart rate (f_H) were strongly correlated with oxygen consumption [Fig. 2(b), (c); $V_{O_2} = 27.0 E_{O_2} + 0.736$, $r^2 = 0.567$, $P < 0.001$, $n = 9$; $V_{O_2} = 0.030 f_H + 0.382$, $r^2 = 0.452$, $P < 0.001$, $n = 9$].

RESPONSE TO A DECREASE IN WATER TEMPERATURE

Oxygen consumption decreased exponentially from 2.5 to 1.3 mg kg⁻¹ min⁻¹ when water temperature was reduced from 21 to 12°C at a rate of 0.5°C h⁻¹ [Fig. 3(a); $V_{O_2} = 0.473 \times 1.082^T$, $r^2 = 0.685$, $P < 0.001$, $n = 9$]. An analysis of covariance performed on log transformed data revealed that the slope of the

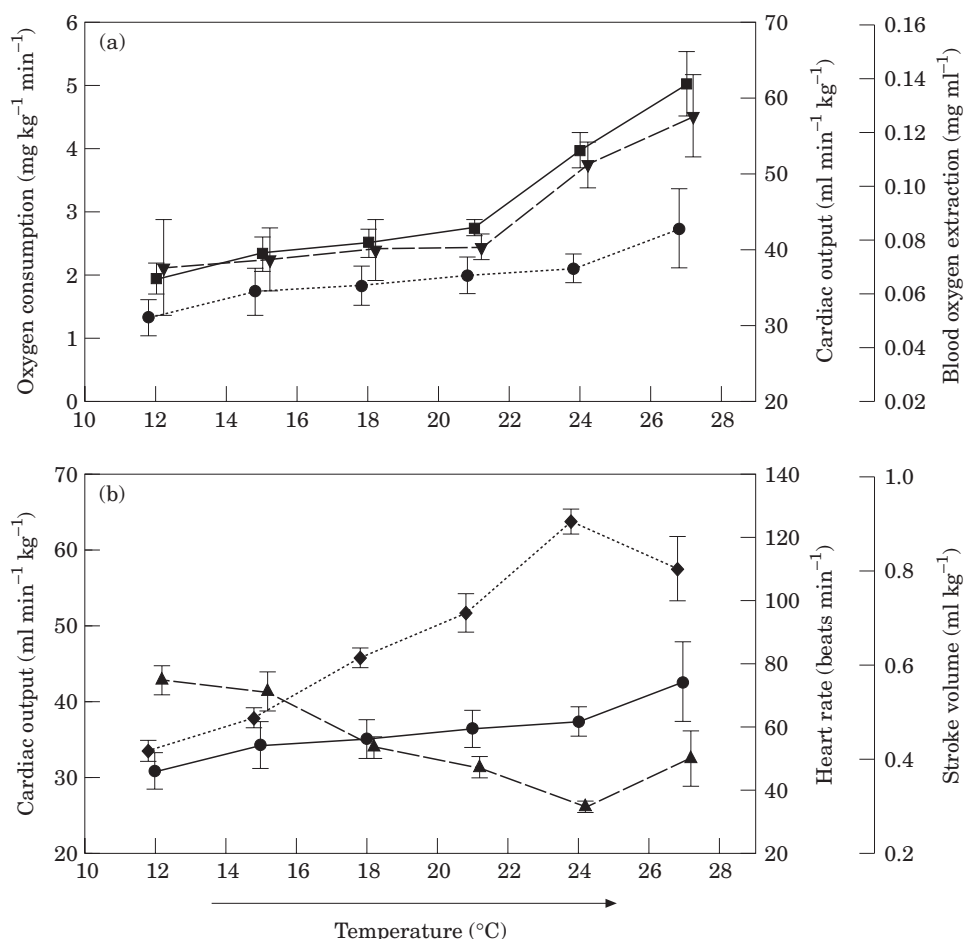


FIG. 1. Cardiovascular adjustments of rainbow trout ($n=9$, mean $\pm$ S.E.) during an acute increase of water temperature at a rate of 2°C h^{-1} . (a) Interaction of cardiac output ($\bullet$) and blood oxygen extraction ($\blacktriangledown$) in the regulation of oxygen consumption ($\blacksquare$). (b) Interaction of heart rate ($\blacklozenge$) and stroke volume ($\blacktriangle$) in the regulation of cardiac output ($\bullet$).

regression line obtained between oxygen consumption and temperature when water temperature was increased did not differ significantly from the slope obtained when temperature decreased ($P=0.492$). However, the intercepts of the two regression lines were significantly different ($P<0.001$). Blood oxygen extraction decreased only slightly as water temperature was reduced, and it is the decrease of cardiac output from 57.2 to $34.3 \text{ ml min}^{-1} \text{kg}^{-1}$ that accounted for $84.3 \pm 14.8\%$ of the decrease in oxygen consumption [Fig. 3(a)]. The reduction of cardiac output was caused by a large decrease of heart rate from 82.9 to $39.3 \text{ beats min}^{-1}$ [Fig. 3(b)]. Stroke volume increased from 0.703 to 0.874 ml kg^{-1} [Fig. 3(b)] but this increase was not sufficient to compensate for the large decrease in heart rate observed. Both cardiac output and heart rate showed a strong linear relationship with oxygen consumption [Fig. 4(a), (c); $V_{\text{O}_2}=0.035 C_{\text{O}}+0.210$, $r^2=0.664$, $P<0.001$, $n=9$; $V_{\text{O}_2}=0.023 f_{\text{H}}+0.482$, $r^2=0.621$, $P<0.001$,

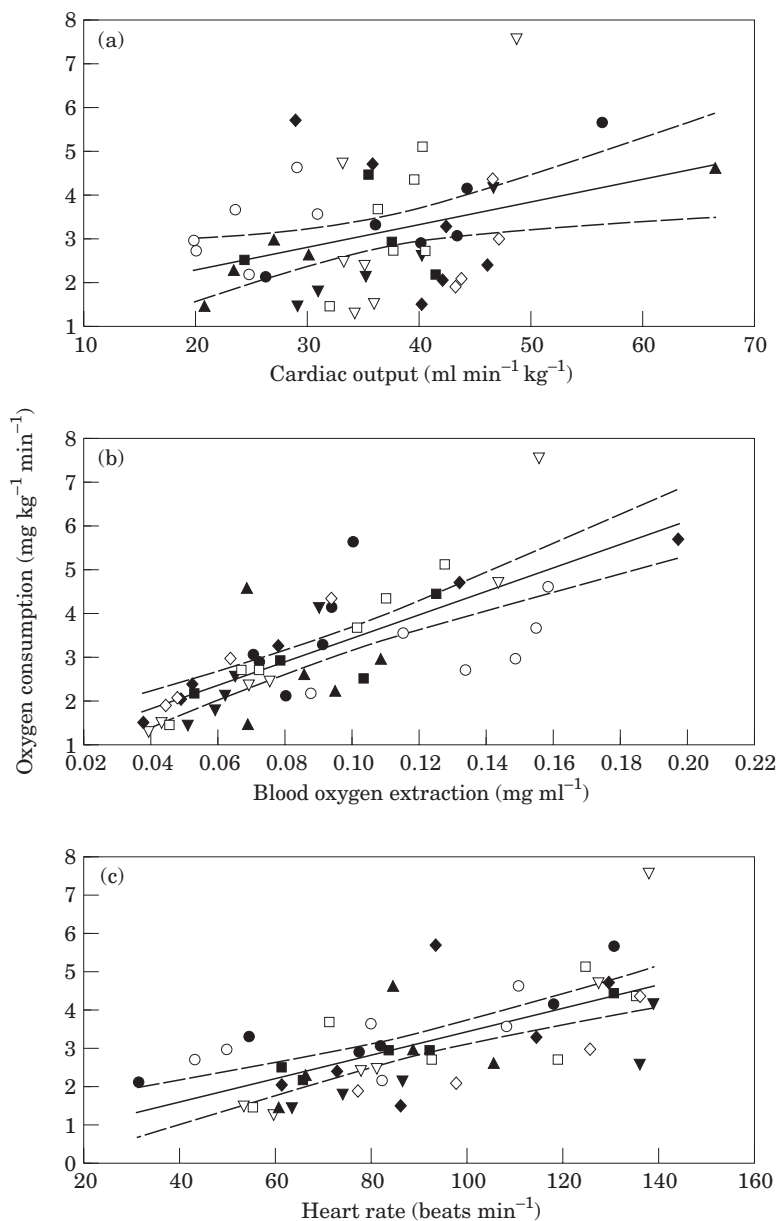


FIG. 2. Relationship between oxygen consumption and (a) cardiac output, (b) blood oxygen extraction and (c) heart rate in rainbow trout submitted to an acute increase of water temperature at a rate of 2° C h⁻¹. Each symbol represents an individual fish. The 95% confidence limits are indicated by the broken lines.

$n=9$], but no correlation was found between blood oxygen extraction and oxygen consumption [Fig. 4(b); $P=0.105$].

RESPONSE TO INCREASING SWIMMING SPEED

The gradual increase in swimming speed (S) caused oxygen consumption to increase exponentially [Fig. 5(a); $V_{O_2}=1.282 \times 3.360^S$, $r^2=0.611$, $P<0.001$, $n=9$].

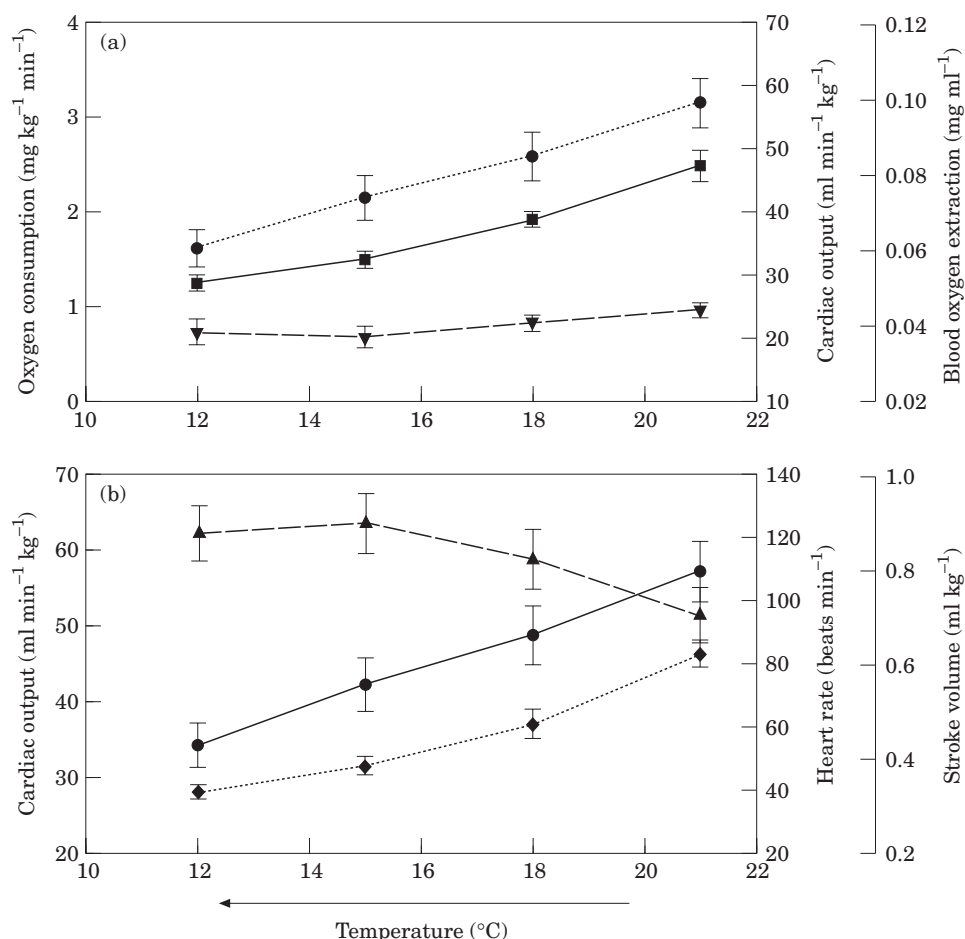


FIG. 3. Cardiovascular adjustments of rainbow trout ($n=9$, mean $\pm$ s.e.) during an acute decrease of water temperature at a rate of $0.5^{\circ}\text{C h}^{-1}$. (a) Interaction of cardiac output ($\bullet$) and blood oxygen extraction ($\blacktriangledown$) in the regulation of oxygen consumption ($\blacksquare$). (b) Interaction of heart rate ($\blacklozenge$) and stroke volume ($\blacktriangle$) in the regulation of cardiac output ($\bullet$).

The increase of blood oxygen extraction accounted for $66.9 \pm 8.4\%$ of the elevation in oxygen consumption while the increase of cardiac output from 45.2 to $66.5 \text{ ml min}^{-1} \text{kg}^{-1}$ accounted for $32.8 \pm 8.4\%$ of the increase [Fig. 5(a)]. The elevation of cardiac output resulted mainly from the increase of heart rate from 50.0 to $72.6 \text{ beats min}^{-1}$ since stroke volume only slightly increased [Fig. 5(b)]. Cardiac output, blood oxygen extraction and heart rate all showed a significant linear relationship with oxygen consumption [Fig. 6(a)–(c); $V_{\text{O}_2} = 0.086 C_{\text{O}} - 0.391$, $r^2 = 0.537$, $P < 0.001$, $n = 9$; $V_{\text{O}_2} = 59.82 E_{\text{O}_2} - 0.186$, $r^2 = 0.517$, $P < 0.001$, $n = 9$; $V_{\text{O}_2} = 0.162 f_{\text{H}} - 4.96$, $r^2 = 0.709$, $P < 0.001$, $n = 9$]. Heart rate was the parameter showing the best correlation coefficient because the regression lines obtained with cardiac output and blood oxygen extraction tended to vary considerably from fish to fish.

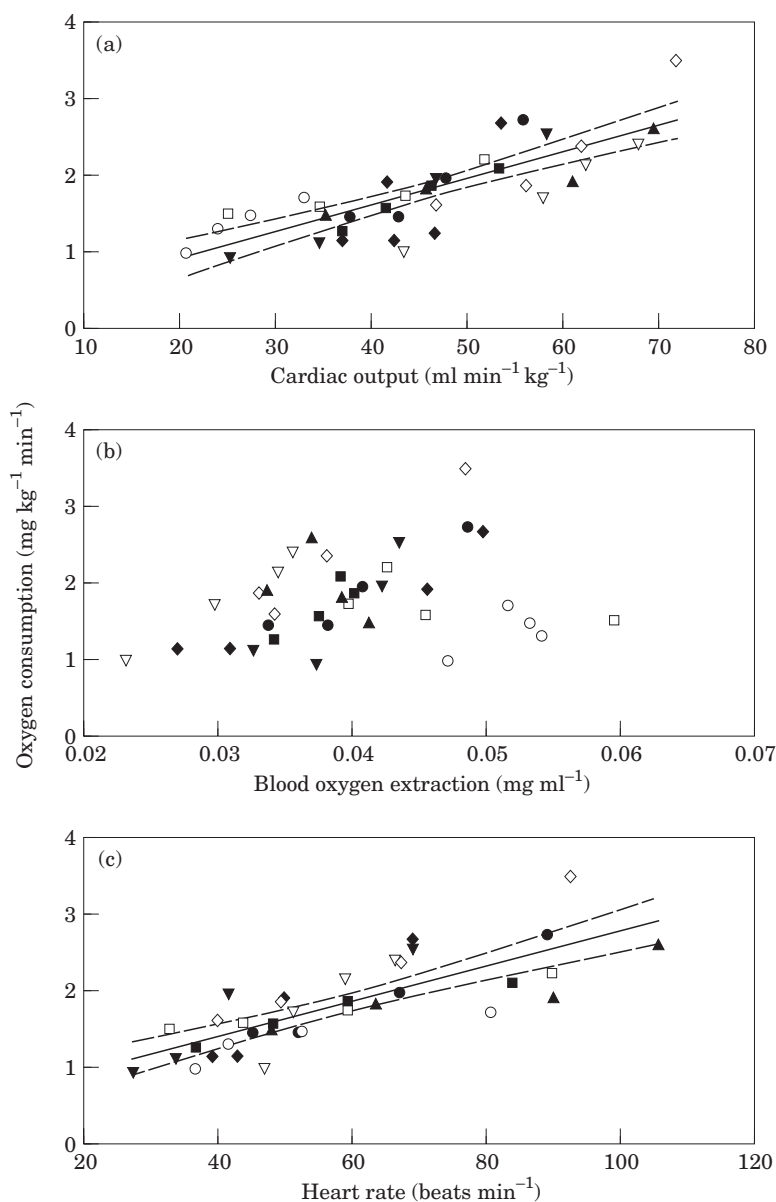


FIG. 4. Relationship between oxygen consumption and (a) cardiac output, (b) blood oxygen extraction and (c) heart rate in rainbow trout submitted to an acute decrease of water temperature at a rate of $0.5^{\circ} \text{C h}^{-1}$. Each symbol represents an individual fish. The 95% confidence limits are indicated by the broken lines.

VARIABILITY OF THE REGRESSION LINES BETWEEN TREATMENTS

For both cardiac output and heart rate, the slope of the regression line with oxygen consumption did not differ significantly when water temperature was either increased or decreased [Fig. 7(a), (c); $P=0.389$ for cardiac output and $P=0.513$ for heart rate]. However, in both cases, the intercept was significantly higher when water temperature was decreased compared with when it was

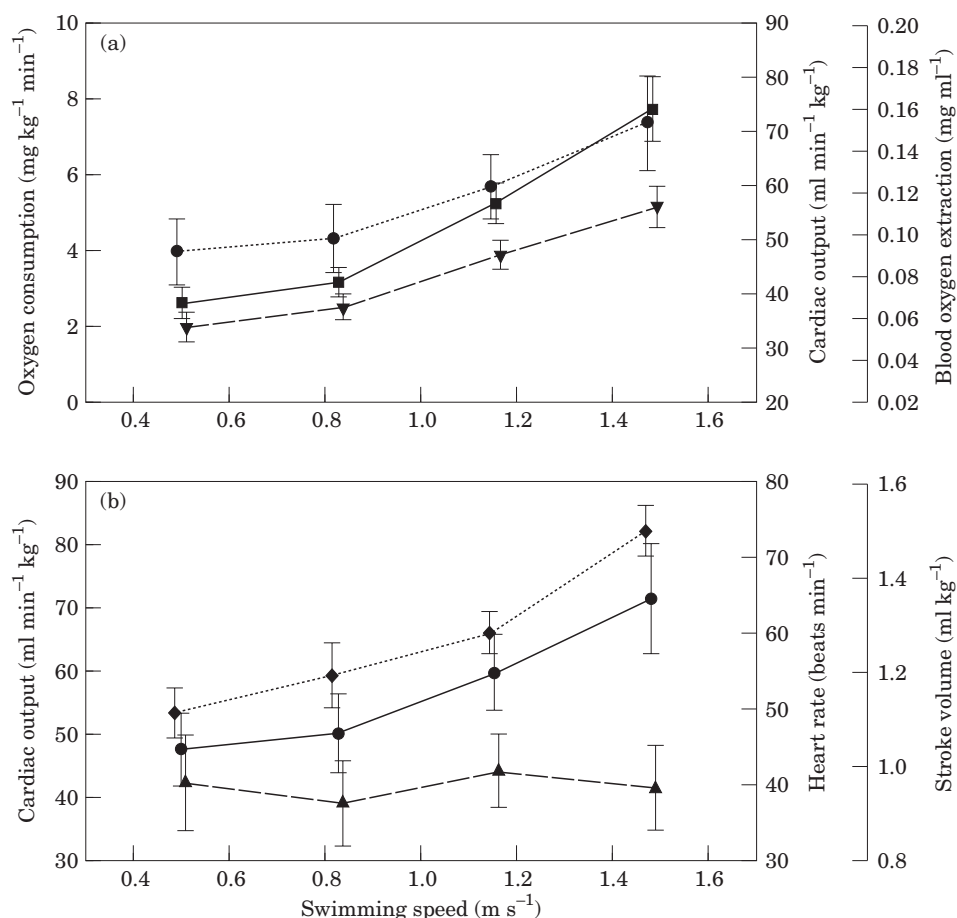


FIG. 5. Cardiovascular adjustments of rainbow trout ($n=9$, mean $\pm$ S.E.) during a gradual increase in swimming speed. (a) Interaction of cardiac output ($\bullet$) and blood oxygen extraction ($\blacktriangledown$) in the regulation of oxygen consumption ($\blacksquare$). (b) Interaction of heart rate ($\blacklozenge$) and stroke volume ($\blacktriangle$) in the regulation of cardiac output ($\bullet$).

increased ($P<0.001$ for both cardiac output and heart rate). Also, for both cardiac output and heart rate, the slope of the regression line with oxygen consumption was elevated significantly more when the fish were forced to swim at increasingly higher swimming speeds than when water temperature was increased or decreased [Fig. 7(a), (c); $P=0.017$ for cardiac output and $P<0.001$ for heart rate]. The slope of the regression line between oxygen consumption and blood oxygen extraction was also significantly higher when the fish were forced to swim than when water temperature was decreased [Fig. 7(b); $P=0.001$].

DISCUSSION

The objective of this study was to determine if cardiac output was a better indicator of metabolic rate than was heart rate. If a cardiovascular variable is to become a reliable predictor of metabolic rate, it needs to satisfy two main requirements. First, its relationship with oxygen consumption must be stable

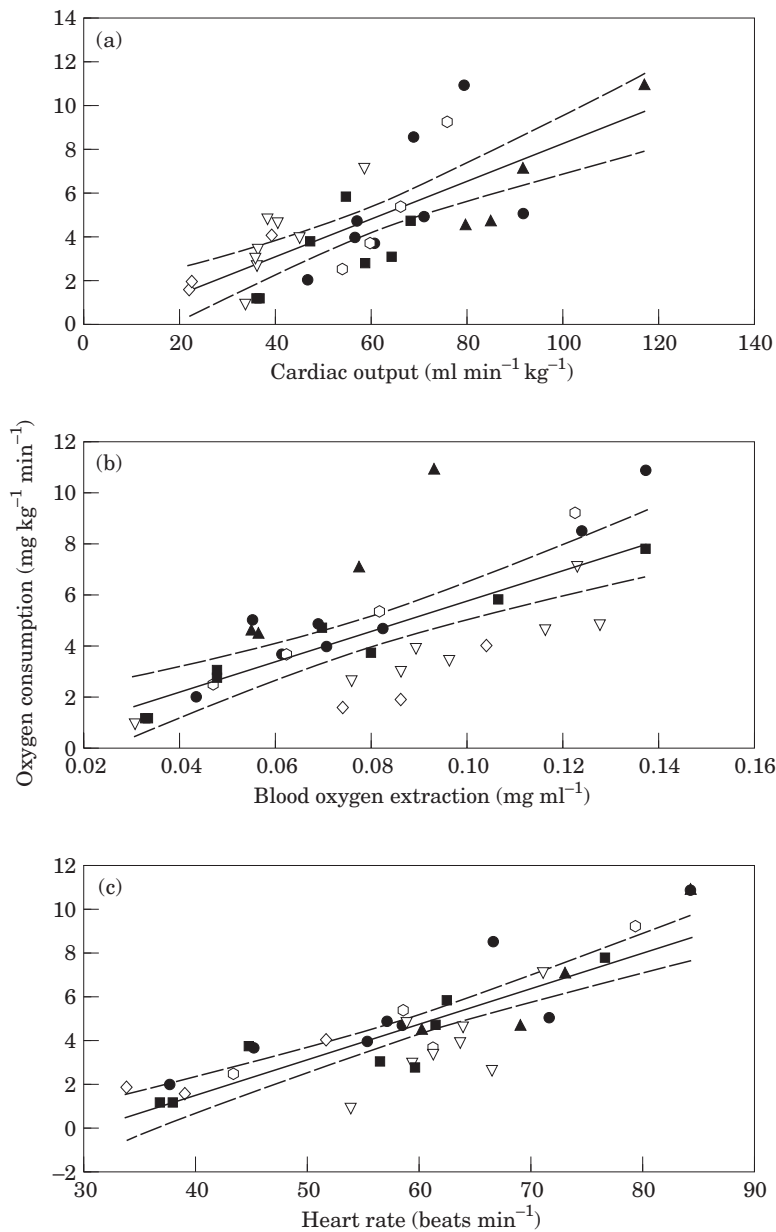


FIG. 6. Relationship between oxygen consumption and (a) cardiac output, (b) blood oxygen extraction and (c) heart rate in rainbow trout submitted to a gradual increase in swimming speed. Each symbol represents an individual fish. The 95% confidence limits are indicated by the broken lines.

under all environmentally relevant conditions so that a single linear (or log-linear) relationship can be established. Secondly, this relationship must present limited variation so that the oxygen consumption value predicted from the regression line is accurate and meaningful (Thorarensen *et al.*, 1996a).

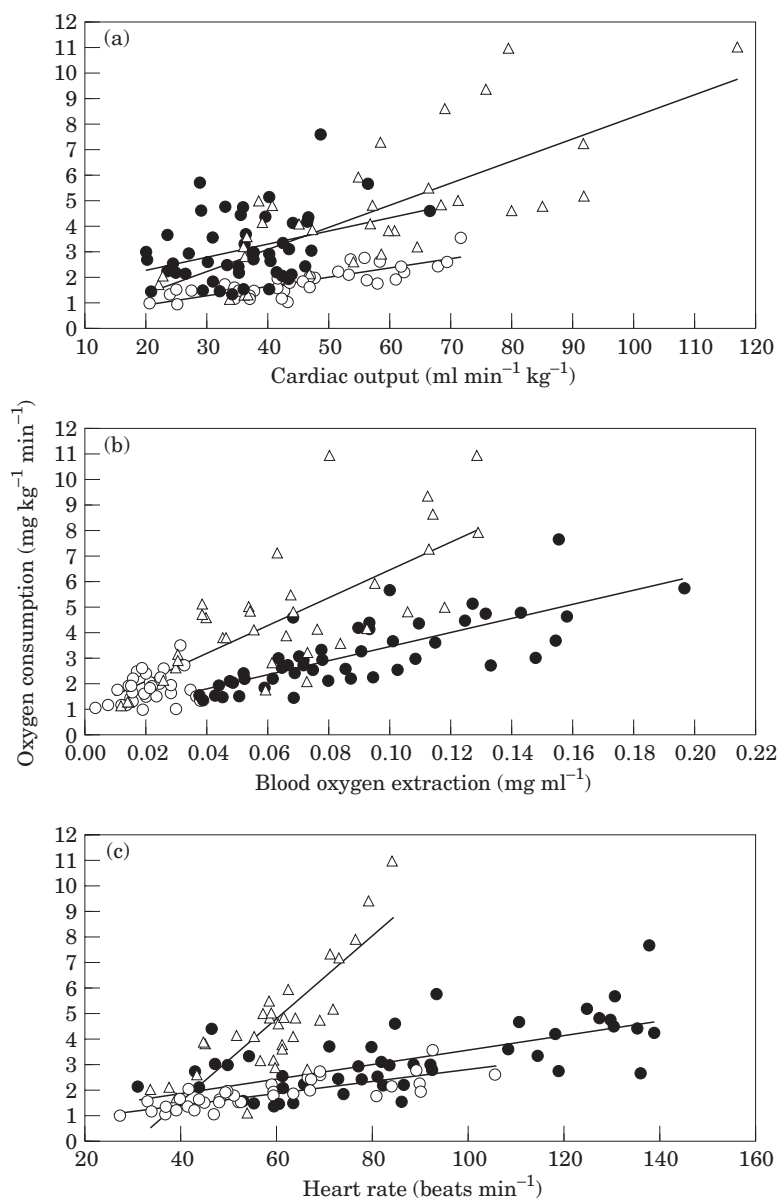


FIG. 7. Relationship between oxygen consumption and (a) cardiac output, (b) blood oxygen extraction and (c) heart rate in rainbow trout subjected to an acute increase of water temperature (●), an acute decrease of water temperature (○) or an increase of swimming speed (△).

This study showed that, in rainbow trout, the relationship between cardiac output and oxygen consumption depends on the factor altering metabolic rate since each treatment resulted in a different regression line [Fig. 7(a)]. For a universal linear relationship to exist between cardiac output and oxygen consumption, blood oxygen extraction would have to be either constant or vary in a predictable manner with cardiac output as oxygen consumption changes. However, this study showed that the contribution of cardiac output and blood

oxygen extraction to the modifications of oxygen consumption varies depending on the circumstances, which is why the relationship between oxygen consumption and cardiac output can be expressed by more than a single regression line.

Cech *et al.* (1976) also observed variation in the contribution of cardiac output and blood oxygen extraction to the elevation of oxygen consumption in winter flounder *Pseudopleuronectes americanus* (Walbaum) acclimated to winter and summer temperatures. The cardiovascular adjustments changed from an increase in blood oxygen extraction with no change in cardiac output at winter temperatures to an increase in cardiac output and relatively unchanged blood oxygen extraction at summer temperatures.

The present study also showed that, although a fairly strong correlation coefficient can at times be found between oxygen consumption and cardiac output [Figs 4(a) and 6(a)] there are situations where the variations in oxygen consumption are related only weakly to cardiac output [Fig. 2(a)]. Furthermore, as the data obtained by increasing swimming speed show [Fig. 6(a)], even if a significant relationship is found between cardiac output and oxygen consumption, the predictive power of the regression line can remain limited considering the substantial variation of the data. Indeed, while the oxygen consumption predicted from the regression line obtained by increasing swimming speed would be $3 \text{ mg kg}^{-1} \text{ min}^{-1}$ for a cardiac output of $40 \text{ ml min}^{-1} \text{ kg}^{-1}$, the actual value could be anywhere between 2.2 and $4 \text{ mg kg}^{-1} \text{ min}^{-1}$ when taking into account the 95% confidence limit of the predicted value.

Thus, the data obtained in this study show that the problems of multiple regression lines and variation highlighted by Thorarensen *et al.* (1996a) about the use of heart rate as a predictor of metabolic rate are also of concern with cardiac output. Often heart rate was correlated even more strongly with oxygen consumption than cardiac output, although the problem of scatter of data points [Fig. 2(c)] and variation of regression lines [Fig. 7(c)] was observed again. A potential source of variation in this experiment is the calibration of the flow probes. Even though great care was always taken during the calibrations, slight changes in the position of the probe could affect the results significantly. It is difficult to estimate the accuracy of the calibrations, but variation at this level could 'muddy' the relationship obtained between cardiac output and oxygen consumption.

A surprising finding was that the elevation of cardiac output during swimming depended almost entirely on an increase in heart rate with almost no variation in stroke volume. This result was unexpected since the relative contribution of stroke volume to the increase of cardiac output is usually considered to be similar to or greater than that of heart rate (Farrell & Jones, 1992). However, the important role of heart rate seen during swimming in this study is consistent with recent evidence suggesting that the contribution of heart rate may have been underestimated in earlier work in which the fish underwent invasive surgery (Thorarensen *et al.*, 1996a; Webber *et al.*, 1998; Altimiras & Larsen, 2000). On the other hand, the possibility for further increase in stroke volume may have been limited by the fact that cardiac output and stroke volume were already quite high before swimming [Fig. 5(b)] compared with basal values observed in other studies (Kiceniuk & Jones, 1977; Taylor *et al.*, 1996; Thorarensen *et al.*, 1996b), or before temperature was increased [Fig. 1(b)].

Although heart rate may contribute significantly to increase cardiac output in some situations, the increase of heart rate appears to be inefficient for elevating cardiac output when it is too severe. Such a phenomenon was seen when water temperature was increased acutely. In this situation, cardiac output increased only moderately because a decrease in stroke volume counteracted the large elevation of heart rate [Fig. 1(b)]. The decrease of stroke volume is the result of a reduction of both filling time and force of contraction (Farrell & Jones, 1992). When water temperature is increased at a slower rate, as was the case before the experiment involving a decrease in water temperature, the increase in heart rate is less severe and cardiac output is higher (37 v. $57 \text{ ml min}^{-1} \text{ kg}^{-1}$ at 21°C during the increase and decrease in water temperature, respectively). Similar severe but ineffective increases in heart rate have been observed also in fish exposed to acutely lethal concentrations of metals and have been proposed to be part of a general acute toxicity syndrome (Wilson & Taylor, 1993; Brodeur *et al.*, 1999). The observation of a similar phenomenon during the acute increase of water temperature suggests that this response may be a general reaction to a wider range of physiological stressors.

In conclusion, this study has shown that cardiac output presents few advantages over heart rate as a predictor of metabolic rate in rainbow trout. The complex nature of the interactions existing between cardiac output, heart rate and blood oxygen extraction in the control of oxygen consumption calls for more research to be done before any cardiac parameter can be used to predict metabolic rate in free-swimming fish. A better understanding of the contributions of cardiac output and heart rate to the variation of oxygen consumption under various environmental conditions eventually might allow prediction of metabolic rate from a set of regression lines, provided there is some information about the nature of the factors affecting metabolic rate. However, the problem of having to calibrate each fish individually may remain if inter-individual variation is always important. Some species other than rainbow trout might be more suited to the use of cardiac output as a predictor of metabolic rate since Webber *et al.* (1998) obtained a good relationship between cardiac output and oxygen consumption in the cod.

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